Luiss
Libera Università Internazionale
degli Studi Sociali Guido Carli

Lists

Introduction to Computer Programming

Management and Artificial Intelligence

LUISS

A List is a Sequence of Items

In Python, lists:

- are ordered **sequences** of items that can be accessed via index (remember: Python is 0-based!)
- are denoted by **square brackets**, i.e. [ ]

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- *homogeneous* (e.g., all integers, all strings, ...)
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A list contains **elements**, that can either be:

- *homogeneous* (e.g., all integers, all strings, ...)
- *heterogeneous* (less common)

Conversely to strings, lists are **mutable**:

- elements can be changed

Indices and Ordering

```
a_list = []                        # an empty list
L = [2, 'a', 4, [1, 2]]            # an heterogeneous list

print(f'len(L) = {len(L)}')
print(f'L[0] = {L[0]}')
print(f'L[2] + 1 = {L[2] + 1}')
print(f'L[3] = {L[3]}')            # another list
```

```
len(L) = 4
L[0] = 2
L[2] + 1 = 5
L[3] = [1, 2]
```

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NOTE: List elements are indexed from `0` to `len(L)-1`.

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```
len(L) = 4
L[0] = 2
L[2] + 1 = 5
L[3] = [1, 2]
```

NOTE: List elements are indexed from `0` to `len(L)-1`.

HINT: Use `range(len(L))` to get the valid indices of a list `L`.

```python
print(L[4])
```

```
---------------------------------------------------------------------------
IndexError                                Traceback (most recent call last)
Cell In[22], line 1
----> 1 print(L[4])

IndexError: list index out of range
```

```python
i = 2
print(f'L[i-1] = {L[i-1]}')
```

```
L[i-1] = a
```

```python
print(f'L[1:3] = {L[1:3]}')
```

```
L[1:3] = ['a', 4]
```

Mutability

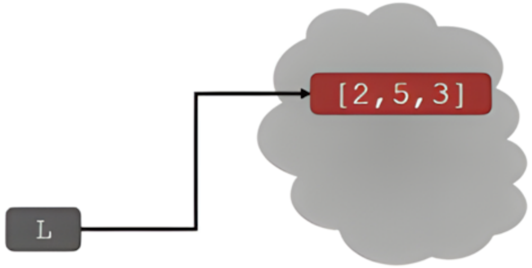
NOTE: Lists are **mutable**.

The assignment of an element at a given index changes the value!

```
L = [2, 1, 3]
print(f'Original L: {L}')
L[1] = 5
print(f'Modified L: {L}')
```

```
Original L: [2, 1, 3]
Modified L: [2, 5, 3]
```

NOTE: This is the **same object** `L`.



Iterating

Common strategies:

1. Iterate over the elements of the list:
    - Simpler: no need to mess with indices
    - We can only read the elements but we cannot update them
2. Iterate over the indices of the list:
    - Slightly harder: we need to use the indices
    - We can both read and update the elements.

Example: compute the sum of elements of a list

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Approach #1: Iterate by element

```
L = [2, 5, 3]
total = 0
for e in L:          # Extract each element of the list
    total += e
print(f'Sum: {total}')
```

```
Sum: 10
```

Example: compute the sum of elements of a list

Approach #1: Iterate by element

```
L = [2, 5, 3]
total = 0
for e in L:         # Extract each element of the list
    total += e
print(f'Sum: {total}')
```

```
Sum: 10
```

Approach #2: Iterate by index

```
L = [2, 5, 3]
total = 0
for i in range(len(L)): # Using range(len(L)) to get valid indexes
    total += L[i]       # extract element at index i
print(f'Sum: {total}')
```

```
Sum: 10
```

Updating elements while iterating

Suppose we want to increment by 1 each element of a list while iterating over it.

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Approach #2: Iterate by index

```
L = [2, 5, 3]
for i in range(len(L)): # Using range(len(L)) to get valid indexes
    L[i] += 1           # we update the original element through indexing
print(f'L: {L}')        # OK: the list has been updated
```

```
L: [3, 6, 4]
```

Updating elements while iterating

Suppose we want to increment by 1 each element of a list while iterating over it.

Approach #2: Iterate by index

```
L = [2, 5, 3]
for i in range(len(L)): # Using range(len(L)) to get valid indexes
    L[i] += 1           # we update the original element through indexing
print(f'L: {L}')        # OK: the list has been updated
```

```
L: [3, 6, 4]
```

Approach #1: Iterate by element

```
L = [2, 5, 3]
for e in L:          # Extract each element of the list
    e += 1           # We only get a copy of the element
print(f'L: {L}')     # KO: the list was NOT updated!
```

```
L: [2, 5, 3]
```

Basic Operations

`append()`

Add elements to the end of a list `L` with:

```
L.append(<elements>)
```

append()

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```
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```

```
L = [2, 1, 3]
print(f'Original L: {L}')
L.append(5) # lists are objects with data, methods and functions
print(f'L after append: {L}')
```

```
Original L: [2, 1, 3]
L after append: [2, 1, 3, 5]
```

`append()`

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L.append(<elements>)
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```
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print(f'L after append: {L}')
```

```
Original L: [2, 1, 3]
L after append: [2, 1, 3, 5]
```

NOTE: `append()` mutates the list!

NOTE: Like methods in strings, methods in lists also use the dot notation.

+ vs. `extend()`

Concatenate lists L1 and L2 together with:

```
L1.extend(L2)
```

or:

```
L1 + L2
```

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```
L1.extend(L2)
```

or:

```
L1 + L2
```

```
L1 = [2, 1, 3]
L2 = [4, 5, 6]
L3 = L1 + L2
print(f'L1: {L1}')
print(f'L2: {L2}')
print(f'L3 = L1 + L2: {L3}')

L1.extend(L2)
print(f'L1 after extend: {L1}')
```

```
L1: [2, 1, 3]
L2: [4, 5, 6]
L3 = L1 + L2: [2, 1, 3, 4, 5, 6]
L1 after extend: [2, 1, 3, 4, 5, 6]
```

Similar but different...

NOTE: `extend()` mutates the list, whereas the `+` operator returns a new list.

`pop()` vs. `remove()` vs. `del()`

You can **remove** item(s) from a list using three different strategies:

- `del()`: delete element at a **specific index** `i` with `del(L[i])`
- `pop()`: delete element at a **specific index** `i` (and return the removed element) with `L.pop(i)`. If index is not provided, `L.pop()` deletes the last element of the list by default.
- `remove()`: remove a **specific element** `x` with `L.remove(x)`.
    - `L.remove(x)` looks for the element `x` in `L` and, if found, removes it
    - if `x` occurs multiple times, removes first occurrence only
    - if `x` is not found in `L`, throws an error

`pop()` vs. `remove()` vs. `del()`

You can **remove** item(s) from a list using three different strategies:

- `del()`: delete element at a **specific index** `i` with `del(L[i])`
- `pop()`: delete element at a **specific index** `i` (and return the removed element) with `L.pop(i)`. If index is not provided, `L.pop()` deletes the last element of the list by default.
- `remove()`: remove a **specific element** `x` with `L.remove(x)`.
  - `L.remove(x)` looks for the element `x` in `L` and, if found, removes it
  - if `x` occurs multiple times, removes first occurrence only
  - if `x` is not found in `L`, throws an error

NOTE: *All* of the three methods **mutate** the list.

```python
L = [2, 1, 3, 6, 3, 7, 0]
print(f'Original L: {L}')

L.remove(2)
print(f'After L.remove(2): {L}')

L.remove(3) # removes first occurrence
print(f'After L.remove(3): {L}')

del(L[1])
print(f'After del(L[1]): {L}')

a = L.pop()
print(f'After a = L.pop(): {L}')
print(f'a = {a}')
```

```
Original L: [2, 1, 3, 6, 3, 7, 0]
After L.remove(2): [1, 3, 6, 3, 7, 0]
After L.remove(3): [1, 6, 3, 7, 0]
After del(L[1]): [1, 3, 7, 0]
After a = L.pop(): [1, 3, 7]
a = 0
```

NOTE:

1. when running `L.remove(3)`, only the first `3` is gone
2. when using `L.pop()`, the assignment of the returned value is not mandatory

WARNING: Never mutate a list as you iterate through it.

Example: write a function that removes duplicates from a list `L1` that are present in `L2`.

```
def remove_dups_buggy(L1, L2):
    for item in L1:
        if item in L2:
            L1.remove(item)
    return L1

L1 = [1, 2, 3, 4]
L2 = [1, 2, 5, 6]
print(f'Result (buggy): {remove_dups_buggy(L1, L2)}')
```

```
Result (buggy): [2, 3, 4]
```

This is incorrect. Why?

- Python uses an internal counter to keep track of the position in the loop
- Mutation changes the length, but Python does not update the counter
- The loop never visits element `2` in `L1`

The correct way:

```python
def remove_dups_correct(L1, L2):
    L1copy = L1[:]
    for item in L1copy:
        if item in L2:
            L1.remove(item)
    return L1

L1 = [1, 2, 3, 4]
L2 = [1, 2, 5, 6]
print(f'Result (correct): {remove_dups_correct(L1, L2)}')
```

```
Result (correct): [3, 4]
```

Strings & Lists

- Convert a string `s` to a list `L` using `L = list(s)`
    - this yields a list where each element is a character from `s`

```
s = "I <3 Python"
L = list(s)
print(f'list(s): {L}')
```

```
list(s): ['I', ' ', '<', '3', ' ', 'P', 'y', 't', 'h', 'o', 'n']
```

- Split a string `s` on a character parameter* using `s.split()` (splits on whitespace if called with no parameters)
    - this yields a list where each element is a portion of `s`

```
L_split = s.split()
print(f's.split(): {L_split}')
```

```
s.split(): ['I', '<3', 'Python']
```

```
L_split_custom = s.split('<')
print(f"s.split('<'): {L_split_custom}")
```

```
s.split('<'): ['I ', '3 Python']
```

* Splits on whitespace if called with no parameters

- Merge a list of characters/strings `L` using `''.join(L)`
  - this yields a string which sees the concatenation of items in `L` with a given character** in-between (you can specify a character between quotes to add between every element)

```
L_join = ['Hello', 'World']
s_join = ''.join(L_join)
print(f"''.join(['Hello', 'World']): {s_join}")
```

```
''.join(['Hello', 'World']): HelloWorld
```

```
s_join_custom = '_'.join(L_join)
print(f"'_'.join(['Hello', 'World']): {s_join_custom}")
```

```
'_'.join(['Hello', 'World']): Hello_World
```

** You can specify a character (between quotes) to add between every element

Other Operations

Python has a set of built-in methods that you can use on lists.

You can find the full list at https://docs.python.org/3/tutorial/datastructures.html

NOTE:

Some list methods return new values. All main list methods mutate the list in-place.

Let's go through the most popular ones, shall we?

- sort elements in a list: `sort()` and `sorted()`
- flip elements last-to-first in a list: `reverse()`

- sort elements in a list: `sort()` and `sorted()`
- flip elements last-to-first in a list: `reverse()`

```
L = [9, 6, 0, 3]
print(f'Original L: {L}')
```

```
Original L: [9, 6, 0, 3]
```

- sort elements in a list: `sort()` and `sorted()`
- flip elements last-to-first in a list: `reverse()`

```
L = [9, 6, 0, 3]
print(f'Original L: {L}')
```

```
Original L: [9, 6, 0, 3]
```

```
L1 = sorted(L) # returns sorted list in L1, does not change L
print(f'L after sorted(L): {L}')
print(f'L1: {L1}')
```

```
L after sorted(L): [9, 6, 0, 3]
L1: [0, 3, 6, 9]
```

- sort elements in a list: `sort()` and `sorted()`
- flip elements last-to-first in a list: `reverse()`

```
L = [9, 6, 0, 3]
print(f'Original L: {L}')
```

```
Original L: [9, 6, 0, 3]
```

```
L1 = sorted(L) # returns sorted list in L1, does not change L
print(f'L after sorted(L): {L}')
print(f'L1: {L1}')
```

```
L after sorted(L): [9, 6, 0, 3]
L1: [0, 3, 6, 9]
```

```
L.sort() # L is now mutated (sorted)
print(f'L after L.sort(): {L}')
```

```
L after L.sort(): [0, 3, 6, 9]
```

- sort elements in a list: `sort()` and `sorted()`
- flip elements last-to-first in a list: `reverse()`

```
L = [9, 6, 0, 3]
print(f'Original L: {L}')
```

```
Original L: [9, 6, 0, 3]
```

```
L1 = sorted(L) # returns sorted list in L1, does not change L
print(f'L after sorted(L): {L}')
print(f'L1: {L1}')
```

```
L after sorted(L): [9, 6, 0, 3]
L1: [0, 3, 6, 9]
```

```
L.sort() # L is now mutated (sorted)
print(f'L after L.sort(): {L}')
```

```
L after L.sort(): [0, 3, 6, 9]
```

```
L.reverse() # L is now mutated (flipped)
print(f'L after L.reverse(): {L}')
```

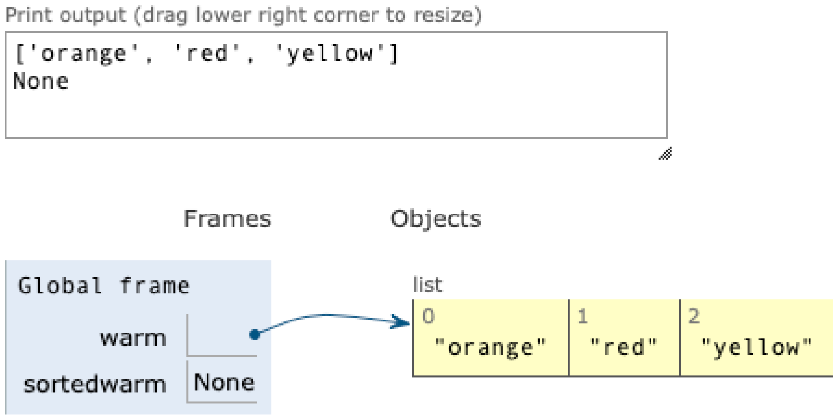
```
L after L.reverse(): [9, 6, 3, 0]
```

To mutate or not to mutate, that is the question.

- calling `sort()` mutates the list, returns nothing (None)
- calling `sorted()` does not mutate the list, must assign to a new variable

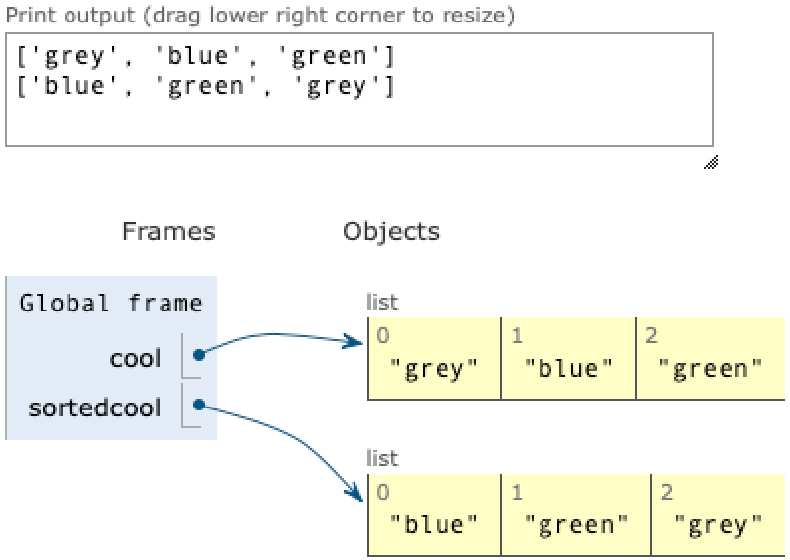
`sort()` mutates:

```
warm = ['red', 'yellow', 'orange']
sortedwarm = warm.sort()
print(f'warm: {warm}')
print(f'sortedwarm: {sortedwarm}')
```



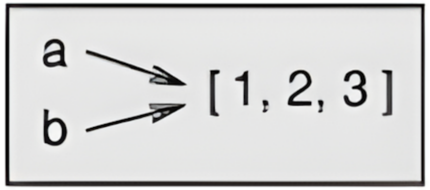
`sorted()` does not mutate:

```
cool = ['gray', 'blue', 'green']
sortedcool = sorted(cool)
print(f'warm: {cool}')
print(f'sortedwarm: {sortedcool}')
```



Aliasing

Aliasing: if `a` refers to an object and you assign `b = a`, then both variables refer to the same object.




```
a = [1, 2, 3]
b = a
print(f'b is a: {b is a}')
```

```
b is a: True
```

- **Reference:** association of a variable with an object (in the example above, two references to the same object)
- **Aliased:** an object with more than one reference has more than one name, so we say that the object is *aliased*.

If the aliased object is mutable (like lists), changes made to one alias affect the other:

```
a = [1, 2, 3]
b = a
b[0] = 42
print(f'b: {b}')
print(f'a: {a}')
```

```
b: [42, 2, 3]
a: [42, 2, 3]
```

WARNING: This aspect can be useful, but **error-prone**!

HINT: To be safe, **avoid aliasing** when working **with mutable objects**.

The aliasing side-effect also holds for list methods we saw earlier:

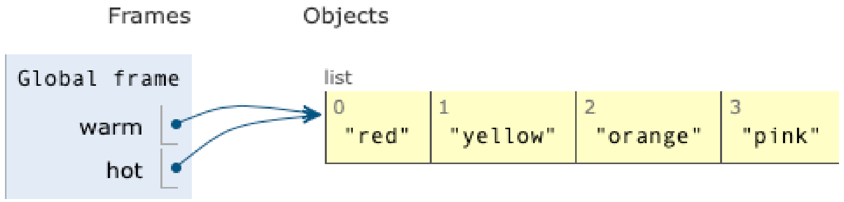
```python
warm = ['red', 'yellow', 'orange']
hot = warm                                  # we have an alias here
hot.append('pink')
print(f'hot: {hot}')
print(f'warm: {warm}')                      # breakpoint #1

warm.sort()
print(f'hot after sort: {hot}')
print(f'warm after sort: {warm}')   # breakpoint #2
```

```
hot: ['red', 'yellow', 'orange', 'pink']
warm: ['red', 'yellow', 'orange', 'pink']
hot after sort: ['orange', 'pink', 'red', 'yellow']
warm after sort: ['orange', 'pink', 'red', 'yellow']
```

Breakpoint #1

Print output (drag lower right corner to resize)

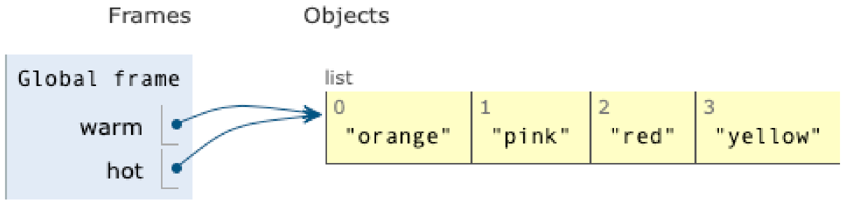
```
['red', 'yellow', 'orange', 'pink']
['red', 'yellow', 'orange', 'pink']
```



Breakpoint #2

Print output (drag lower right corner to resize)

```
['red', 'yellow', 'orange', 'pink']
['red', 'yellow', 'orange', 'pink']
['orange', 'pink', 'red', 'yellow']
['orange', 'pink', 'red', 'yellow']
```



Aliasing: Mutable vs Immutable Objects

Immutable data types, such as int, float, str, and tuple, are never aliased:

```python
a = 1000
b = a                            # this is NOT an alias
b -= 1                           # this operation does not affect a
print("Original value:", a)
print("Modified value:", b)
```

```
Original value: 1000
Modified value: 999
```

Cloning

Cloning: create a new list (different object) by copying every element via assignment `b = a[:]`

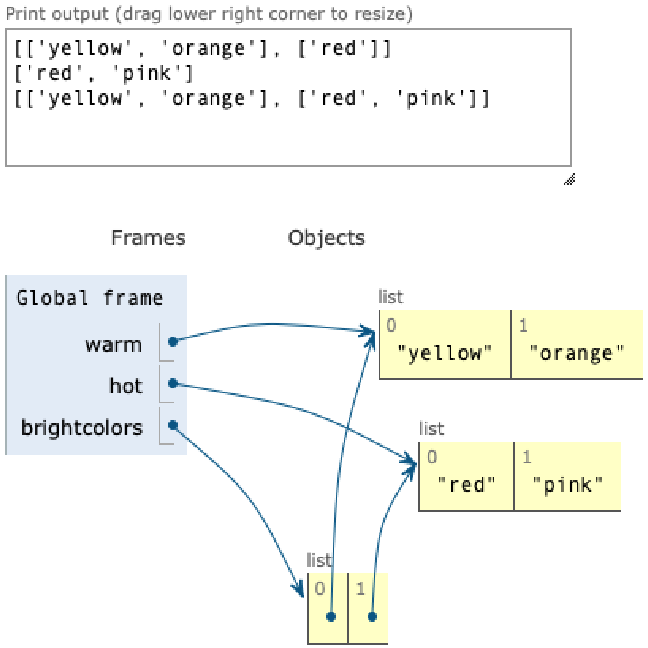
```
cool = ['blue', 'green', 'grey']
chill = cool[:]  # we have a clone here
chill.append('black')
print(f'cool: {cool}')
print(f'chill: {chill}')
```

```
cool: ['blue', 'green', 'grey']
chill: ['blue', 'green', 'grey', 'black']
```

Nesting

RECALL: Lists can be heterogeneous. An element of a list can itself be a list.

WARNING: Mutation and aliasing affect also nested lists.




```python
warm = ['yellow', 'orange']
hot = ['red']
brightcolors = [warm]
brightcolors.append(hot) # brightcolors: [['yellow', 'orange'], ['red']]
hot.append('pink') # hot: ['red', 'pink']
```